Cole Edmonston

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EDUCATION

California Polytechnic State University, San Luis Obispo — **San Luis Obispo, CA**

B.S. in Computer Science — *Expected Graduation:* ***December 2028***

GPA: 3.82 (President's List)

Relevant Coursework: Computer Security, Software Engineering, Systems Programming, Data Structures, Fundamentals of Computer Science, Linear Analysis, Computer Organization

EXPERIENCE

Mustang Market | TypeScript, React, Next.js, Expo, Firebase — **San Luis Obispo, CA**

Co-Founder & Full Stack Developer — March 2026 – Present

- Launched a Cal Poly student marketplace, growing to 1,000+ users (400+ app downloads) via campus events and campaigns
- Engineered a listing pipeline integrating image uploads, Groq Vision auto-tagging, and input validation across 100+ listings
- Integrated Azure AD SSO with Cal Poly's identity provider to authenticate and gate the marketplace to verified students

Spread Goodness | JavaScript, React, Playwright, Docker, Firebase — **Remote (Santa Barbara, CA)**

Software Engineering Intern — May 2026 – September 2026

- Built Playwright regression suites for two critical user flows, automatically validating pull requests from 20+ developers
- Designed seed data generation to simulate realistic user states, enabling consistent, repeatable test runs
- Developed reusable authentication fixtures for six permission levels, eliminating redundant logins across Playwright test runs

PROJECTS

Catan Board Visualizer | Python, PyTorch, YOLO11, OpenCV

- Built a CV referee using YOLO11 and homography to read and regenerate a Catan board from one photo and score players
- Labeled 3,100+ pieces and trained 5 YOLO11 models; cut false detections 43% by targeting lighting where the model failed
- Raised number-token accuracy from 80% to 100% by catching the model reading terrain, not digits, and masking it out

Vision Guided UAV (Simulation) | Python, PX4, ROS 2, Gazebo, OpenCV, MAVSDK

- Built a vision-guided autonomy stack in PX4 SITL and Gazebo; closed-loop visual position hold in progress
- Developed real-time ArUco detection on a ROS 2 camera feed at 1.4-3.1 ms per 1280x960 frame, 99% detection at 9.4 m
- Traced a 4x camera frame-rate drop to UDP fragmentation; tuned kernel socket buffers to lift it from 6 to 24 Hz

PII Cypher | Python, FastAPI, PyMuPDF, pdfplumber

- Built a reversible PII tokenizer for tax PDFs, encrypting values under per-job AES-256-GCM keys, restorable on demand
- Benchmarked against 54 synthetic docs with 536 labeled identifiers across 6 PII types: 100% precision, 99% recall
- Verified redactions by comparing rasterized pages pixel by pixel at 150 DPI, catching failures text extraction misses

ENVISION Wardrobe | React Native, Google Cloud, Docker, Expo

- Built an outfit engine scoring color harmony, pattern, clothing-fit, and weather, for 90+ TestFlight users
- Integrated Groq Vision API to detect clothing type, colors, patterns, brands, and upload items to a user's wardrobe from a photo
- Self-hosted background removal on Google Cloud using rembg and Docker, eliminating API costs

SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, C#, C, SQL

Frameworks & Libraries: PyTorch, OpenCV, Ultralytics YOLO, React, React Native, Next.js, Node.js, FastAPI, Expo, Tailwind CSS, Playwright, Pandas

Tools & Platforms: Git, GitHub Actions, Docker, ROS 2, PX4, Gazebo, AWS, Google Cloud, Firebase, Supabase, PostgreSQL